WORKED SOLUTION

STEP I 1995, Question 6

Pure • Bernoulli Differential Equations

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s1-q6

The key idea

Neither equation is linear in y, but each is linear in a suitable power of $1/y$. The left-hand side contains y'/y^2, which is (up to sign) the derivative of $1/y$; so writing $u = 1/y$ turns the equation into a first-order linear equation for u, which an integrating factor finishes off. In part (ii) the term y'/y^3 suggests looking at $1/y^2$ instead.

Solution

Part (i)

Put $u = 1/y$. By the chain rule,

$$\frac{\mathrm{d}u}{\mathrm{d}x} = -\frac{1}{y^2}\frac{\mathrm{d}y}{\mathrm{d}x},$$

so the first term of the equation is exactly $-\mathrm{d}u/\mathrm{d}x$, and the second is u. The equation becomes

$$-\frac{\mathrm{d}u}{\mathrm{d}x} + u = \mathrm{e}^{2x}, \qquad \text{that is,} \qquad \frac{\mathrm{d}u}{\mathrm{d}x} - u = -\mathrm{e}^{2x}.$$

This is linear in u. The integrating factor is $\mathrm{e}^{\int -1\,\mathrm{d}x} = \mathrm{e}^{-x}$, and multiplying through gives

$$\frac{\mathrm{d}}{\mathrm{d}x}(u\,\mathrm{e}^{-x}) = \mathrm{e}^{-x}\frac{\mathrm{d}u}{\mathrm{d}x} - \mathrm{e}^{-x}u = -\mathrm{e}^{x}.$$

Integrating, with A an arbitrary constant,

$$u\,\mathrm{e}^{-x} = A - \mathrm{e}^{x}, \qquad \text{so} \qquad u = A\mathrm{e}^{x} - \mathrm{e}^{2x}.$$

Since $y = 1/u$,

$$y = \frac{1}{A\mathrm{e}^{x} - \mathrm{e}^{2x}},$$

which is the required general solution. (The original equation involves $1/y$, so $y \neq 0$ throughout and the substitution loses no solutions; the formula is valid on any interval where $A\mathrm{e}^{x} \neq \mathrm{e}^{2x}$, i.e. avoiding $x = \ln A$ when $A > 0$.)

Part (ii)

Ask yourself Which substitution plays the role of $u = 1/y$ here?

We want the derivative of the new variable to produce y'/y^3. Since

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{1}{y^2}\right) = -\frac{2}{y^3}\frac{\mathrm{d}y}{\mathrm{d}x},$$

the right choice is $v = 1/y^2$. Then the first term is $-\frac{1}{2}\,\mathrm{d}v/\mathrm{d}x$ and the second is v, so

$$-\frac{1}{2}\frac{\mathrm{d}v}{\mathrm{d}x} + v = \mathrm{e}^{2x}, \qquad \text{that is,} \qquad \frac{\mathrm{d}v}{\mathrm{d}x} - 2v = -2\mathrm{e}^{2x}.$$

Now the integrating factor is e^{-2x}:

$$\frac{\mathrm{d}}{\mathrm{d}x}(v\,\mathrm{e}^{-2x}) = -2\mathrm{e}^{2x}\cdot\mathrm{e}^{-2x} = -2.$$

Integrating, with B an arbitrary constant,

$$v\,\mathrm{e}^{-2x} = B - 2x, \qquad \text{so} \qquad \frac{1}{y^2} = (B-2x)\,\mathrm{e}^{2x}.$$

Hence

$$y^2 = \frac{\mathrm{e}^{-2x}}{B-2x}, \qquad y = \pm\frac{\mathrm{e}^{-x}}{\sqrt{B-2x}}.$$

Because $1/y^2 > 0$, we need $B - 2x > 0$: the solution exists only for $x < \frac{1}{2}B$, and the sign is fixed by whichever sign y has initially (it can't change sign, since y is never zero). Unlike part (i), the answer isn't a difference of two exponentials: here e^{2x} on the right exactly cancels the integrating factor e^{-2x}, leaving a constant to integrate, which is why the linear term $-2x$ appears.

Checks

- Substitute back in (ii): with $1/y^2 = (B-2x)\mathrm{e}^{2x}$, differentiating gives $-2y'/y^3 = -2\mathrm{e}^{2x} + 2(B-2x)\mathrm{e}^{2x}$, so $y'/y^3 = \mathrm{e}^{2x} - 1/y^2$, which is the equation.
- Numerically: taking $y(0) = 1$ in (ii) gives $B = 1$, so the formula predicts $y(0.4) = \mathrm{e}^{-0.4}/\sqrt{0.2} \approx 1.49888$. Stepping the differential equation from $x = 0$ numerically gives the same value to ten decimal places. Both general solutions also satisfy their equations at many random points and constants.

Common slips

- Forgetting the minus sign (or the factor 2 in part (ii)) in $\mathrm{d}u/\mathrm{d}x$, which wrecks the linear equation.
- In part (ii), writing only the positive square root, or not noticing that the solution requires $x < \frac{1}{2}B$.

Beyond STEP

Both equations are examples of Bernoulli equations, $y' + P(x)y = Q(x)y^n$: dividing by y^n and substituting $u = y^{1-n}$ always produces a linear equation. Here part (i) is $n = 2$ and part (ii) is $n = 3$. The fact that solutions can "blow up" at a finite x (as in $x \to \frac{1}{2}B$) is a typical feature of nonlinear equations, studied in a first course on ODEs; see Boyce and DiPrima, *Elementary Differential Equations*.

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